

Study Garden

Website Summary

Live URL: <https://shinon-study-pre-website-4b4066c4.onbld.com/>

Overview

Study Garden is a study companion web app built with Next.js 16, React 19, Tailwind CSS, and Firebase (Auth + Firestore). It features AI-powered study tools, social features, and a Japan-inspired aesthetic with a lo-fi music soundscape.

Home

- Monthly calendar with leading/trailing dates, event creation, Google Calendar sync
- Vertical calendar (weekly timeline) with events
- Today's Timeline showing events + lecture notes tagged to today's date
- Blue dots on calendar days that have lecture notes
- Add/edit/delete events with Firestore sync

Notes

- Create notes with title, category, lecture date, content
- File uploads: PDF (inline viewer), PowerPoint, Video files, YouTube URLs
- Inline editing with auto-save (1.5s debounce to localStorage + Firestore)
- KaTeX math rendering + Mermaid graph support
- AI Anki Card generation (interactive flashcards with 3D flip animation)
- AI Quiz generation (interactive multiple-choice with correct/wrong feedback)
- Category management (create/edit/delete custom categories)
- View mode (split-screen) / Edit mode toggle
 - Left (2/3): PDF/Materials + Note Content
 - Right (1/3): Anki Cards + Quiz

Tasks

- Task tracker with title, content, deadline fields
- Auto-categorization: Today / Overdue / Next 7 Days (grouped by date)
- Add/edit/delete tasks with Firestore sync
- Clear completed tasks

Friend (SNS-style)

- Profile header with cover, avatar, follower/following counts
- Study statistics (focus time, sessions, points) synced from Firestore
- Friend search by name/email
- Friend request system (send/accept/decline) via Firestore
- Privacy settings: toggle sharing Today's Tasks and Schedule
- 3-dot menu for privacy toggles
- View friend's profile: study stats, today's tasks, schedule (respects privacy)
- Disconnect/unfollow buttons

Search

- Full-text search across notes, tasks, events, and friends
- Auto-suggestions, categorized results, highlighted matches

AI Features (ai& API — [zai-org/glm-5.2](https://github.com/zai-org/glm-5.2))

- Fox Sensei chatbot — conversational assistant with app data context
 - Markdown rendering, function calling (creates tasks and notes via chat)
 - Anti-hallucination: strictly uses provided app data
- Anki Card generation — AI generates flashcards from note content + files

- Quiz generation — AI generates multiple-choice questions

Audio (Google Gemini — gemini-2.5-flash-lite)

- Audio summarization — Record audio in note editor, transcribe + summarize

Focus Session

- Customizable timer (5-120 minutes)
- Points system (10 points per session)
- Garden stage progression (5 stages)
- Japan-inspired lo-fi music — Web Audio API jazz beat
 - ii-V-I chord progression, swing drums, vinyl crackle, tape warble
- Global persistence — timer and music continue across page navigation
- Widget in header bar (timer, play/pause, reset, ambient toggle)

Auth & Data

- Firebase Auth with session isolation (prevents cross-account data mixing)
- Profile storage — per-user localStorage + Firestore
 - avatarUrl stays local to avoid Firestore 1MB document limit
- Firestore collections: users, notes, tasks, events, friend_requests
- URL hash-based routing (survives refresh)
- Per-user profile storage keys to prevent cross-account contamination

Design

- Japan-inspired color palette (moss green, gold, ivory)
- Fox Sensei character as chatbot icon
- Custom app icon as favicon and sidebar logo
- Responsive layout with collapsible sidebar
- Language toggle (English/Japanese)

Tech Stack

- Frontend: Next.js 16, React 19, Tailwind CSS
- Backend: Next.js API Routes (Node.js)
- Database: Firebase Firestore
- Auth: Firebase Authentication
- AI: ai& API (zai-org/glm-5.2), Google Gemini (audio)
- Music: Web Audio API (custom lo-fi generator)
- Math: KaTeX + react-katex
- Graphs: Mermaid.js
- Deployment: bld (antimony build platform)